Venkata Lakshmi Parimala Pasupuleti

Senior Data + AI Engineer · Python · PHP · ETL/ELT · Schema Design · RAG · LLM Evaluation · AWS · Real Estate Finance · Production AI

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Virginia, USA — Remote

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Hiring Team
Juniper Square — Engineering
Re: Senior Software Engineer, Data (AI)

COVER LETTER

Juniper Square serves the institutional investors who own real estate portfolios. For 8 years at Entrata, I built the data systems those portfolios run on — owner financial reporting, distribution analytics, HAP allocation, HUD payment processing, income eligibility, and rent schedule pipelines across 20+ production modules serving 100K+ monthly active users. Your clients are on one side of the real estate ledger. I spent 8 years building the other side. The financial schemas, the regulatory data models, the compliance pipelines — I know them from inside the stack.

On data engineering at scale: I processed **1.6M HUD and Tax Credit income limit records** via production pipelines (Python, PostgreSQL, PL/SQL), designed schema normalization and data validation layers across all client environments, optimized 200+ SQL queries (30% performance gain through execution plan tuning, indexing, partitioning, and audit tables), led database architecture decisions for complex production schemas, and migrated thousands of client documents to AWS S3 — resolving 90% of legacy space constraints. I also created **56 state-specific housing certification documents** integrated with S3 storage. ETL/ELT pipelines on regulated financial data at production scale is not adjacent experience. It's the job I've been doing.

On AI output evaluation — which your JD lists as preferred: **Leasel** achieves 93.6% precision on 1,240 residential lease PDFs with a deterministic rules engine (fully auditable outputs), a pass@k eval harness, **cross-model comparison across multiple LLM providers, and human-in-the-loop review gates**. That is the exact evaluation framework stack your JD describes — deterministic checks, cross-model comparison, human-in-the-loop — shipped and benchmarked, not on a whiteboard. **AgentProbe** adds a 5-metric LLM-as-judge eval engine with CI/CD regression gates that fire on every commit and a React observability dashboard tracking retrieval drift and latency without human query.

The combination that's hard to find elsewhere: production data pipeline engineering depth on real estate financial data + shipped AI evaluation frameworks + agentic development experience. I can own features end-to-end from the data layer through the LLM layer, and I can measure whether the AI is actually working.

Sincerely,

Venkata Lakshmi Parimala Pasupuleti